BPM2_NB_ESOFT QUIZ 3_S2026

# of Questions: 30

Question #: 1

A 46-year-old man is brought to the emergency department following a rescue after one week at sea. He is severely dehydrated and has a loss of vision in the right eye. Physical examination of the eye shows a retinal detachment. This condition is characterized by the separation of which of the following layers?

A. Choroid from sclera
✓B. Retinal pigment epithelium from photoreceptors
C. Retinal pigment epithelium from choroid
D. Inner nuclear layer from the outer plexiform layer
E. Ganglion cell layer from the inner nuclear layer

Rationale: Retinal detachment is commonly caused by trauma, or reduction in the water content of the vitreous humor second to dehydration (which occurred in this case). Retinal detachment usually occur between the retinal pigment epithelium (RPE) and the layers of the outer segments of the photoreceptors. This is because of the embryological origin of both these layers (a potential space exists in the retina as a trace of the spce between the surfaces of the inner and outer epithelial layers of the optic cup). Therefore, they are not firmly held against each other (the RPE cells are not in direct contact with the rods and cones), and thus is easily prone to separation.

Item Description: HCB.1354 - Retinal detachment

Question #: 2

A 35-year-old man comes to the physician because of infertility concerns. Physical examination shows no significant abnormalities. Semen analysis shows low levels of fructose. Which of the following structures or glands is most likely affected in this case?

A. Prostate gland
✓B. Seminal vesicle
C. Bulbourethral gland
D. Seminiferous tubule
E. Glands of Littré

Rationale: The seminal vesicles secrete fructose. The bulbourethral gland is also called the Cowper's gland, and secretes a very thick mucus. It is stimulated mostly in sexual arousal to release its mucoid secretions. The seminiferous tubules does not secrete fructose. The prostate gland releases PSA, citric acid. Glands of Littre are found alongside the spongy urethra and secrete mucus.

Item Description: HCB.1098 - Seminal vesicles

Question #: 3

A 41-year-old with a close-cropped hairstyle routinely cuts his own hair at home. On one such occasion he sustains a deep cut to the scalp. An infection occurred at the site of the cut which he self-managed and eventually the infection seemingly subsided. Twenty days later, he woke up with a severe headache and nausea. His boyfriend brought him to the emergency department where the on-call resident diagnosed an infection of the superior sagittal sinus. Which of the following veins was most likely responsible for the patient’s current infection?

✓A. Emissary
B. Inferior sagittal
C. Diploic

D. Basilar
E. Intercavernous

Rationale: Correct Answer: A
The emissary veins connect the extracranial venous system with the intracranial venous sinuses. They connect the veins outside the cranium to the venous sinuses inside the cranium. They drain from the scalp, through the skull, into the larger meningeal veins and dural venous sinuses exactly the superior sagittal sinus.

Incorrect Answer:
The diploic veins are large, thin-walled valveless veins that channel in the diploë between the inner and outer layers of the cortical bone in the skull.
The basilar artery supply the brain stem mainly the pons.

Item Description: Emissary veins

Question #: 4

A 35-year-old man worker comes to the emergency department because of sudden onset of pain in his left eye. He says he was using a grinding tool without safety goggles. Physical examination shows a small sharp fragment embedded in the lower portion of his eyeball. Branches of which of the following nerves most likely carry the pain sensation from the conjunctiva to the CNS?

A. Optic
B. Maxillary
✓C. Ophthalmic
D. Glossopharyngeal
E. Facial

Rationale: The conjunctiva is innervated by the ophthalmic branch of the trigeminal nerve.

Item Description: A 35-year-old man wo

Question #: 5

A 64-year-old woman is diagnosed with a brainstem tumor. One of her symptoms is an inability to look inwards with her right eye. Which of the following additional symptoms is most likely in her right eye?

✓A. Dilated pupil
B. Inability to look outwards
C. Inability to close the eye
D. Pupil constricted

Rationale: Inability to look inward with the right eye suggests damage to the oculomotor nerve. The oculomotor nerve is also responsible for constricting the pupil in response to light.

Item Description: A 64-year-old woman

Question #: 6

A 1-year-old boy is brought to the physician by his mother because of a 1-day history of incessant crying and distress. A diagnosis of a middle ear infection is made. Which of the following nerves most likely carries the pain sensation from the affected area to the CNS?

A. Cochlear
B. Greater petrosal
✓C. Glossopharyngeal
D. Trigeminal
E. Chorda tympani

Rationale: The glossopharyngeal nerve forms the tympanic plexus in the middle ear which carries sensation from this area.

Item Description: A 1-year-old boy is

Question #: 7

A 27-year-old woman comes to the physician because of a 3-month history of deteriorating vision. Physical examination shows both the direct and consensual pupilary light reflexes are normally responsive. Which of the following statements best characterizes the visual components of this reflex?

✓A. The optic nerves travel between the chiasm and eyeball
B. The optic tracts travel between the eyeball and the thalamus
C. The optic nerves travel between the chiasm and the thalamus
D. The optic tracts join together to form the chiasm

Rationale: The optic nerve is formed by the axons of the cells in the retina converging and exiting the posterior aspect of the eyeball. The left and right optic nerves travel from the back of the eyeball through the optic canal and combine to form the optic chiasm. At the chiasm some fibers cross to the contralateral side. From the optic chiasm the optic tracts travel to the thalamus. From the thalamus the information is transmitted to the cortex.

Item Description: A 27-year-old woman

Question #: 8

A first year medical student is examining the cranial nerves on a standardized patient. He places a sweet tasting substance on the anterior tip of the tongue. Which of the following ganglia most likely contains the cell bodies of the special taste fibers from this part of the tongue?

A. Glossopharyngeal
✓B. Geniculate
C. Pterygopalatine
D. Trigeminal
E. Submandibular

Rationale: The taste to the anterior 2/3 of the tongue is mediated by the chorda tympani a branch of the facial nerve. The cell bodies of the chorda tympani is located in the geniculate ganglion of the facial nerve.

Item Description: A first year medical

Question #: 9

A 6-year-old boy develops a painful ulcer after accidentally biting the inner surface of his cheek while eating. Which of the following nerves transmits the pain sensation to the CNS?

A. Lingual
B. Inferior alveolar
C. Middle superior alveolar
D. Buccal branch of VII
✓E. Buccal nerve from V3

Rationale: The cheek is innervated by the buccal branch of the mandibular nerve.

Item Description: A 6-year-old boy dev

Question #: 10

A 36-year-old woman comes to the physician because of a 2-week history of a change in her voice. Physical examination shows inability to abduct one of her vocal cords. Which of the following muscles has most likely been paralyzed?

A. Cricothyroid
B. Transverse arytenoid
C. Oblique arytenoid
D. Thyroarytenoid
✓E. Posterior cricoarytenoid

Rationale: The posterior cricoarytenoid muscle is the only muscle that abducts the larynx. It is innervated by the recurrent laryngeal nerve.

Item Description: A 36-year-old woman

Question #: 11

A 25-year-old man comes to the physician because of a 2-day history of vision changes after striking his head on the steering wheel during a car crash. During physical examination, he is asked to look inward then upward. The integrity of which of the following extraocular muscles is most likely being tested?

A. Superior oblique

B. Lateral rectus
C. Inferior rectus
✓D. Inferior oblique
E. Superior rectus

Rationale: When testing the extraocular muscles one needs to isolate their functions. The inferior oblique is tested by asking the patient to follow the physicians finger medially and upward.

Item Description: A 25-year-old man co

Question #: 12

A 53-year-old woman comes to the physician because she is has blurred vision. Past medical history shows controlled diabetes type II. Imaging studies shows parasympathetic (ps) fibers traveling along the right CN III are affected. Which of the following symptoms will most likely be present in this patient?

✓A. Loss of consensual light reflex when a light is shone in the left eye
B. Right upper eyelid ptosis
C. Right pseudoptosis
D. Constricted pupil in right eye
E. Loss of the right cornea reflex

Rationale: Optic tract carries visual information from both eyes and the pretectal area projects bilaterally to both Edinger-Westphal nuclei: Consequently, the normal pupillary response to light is consensual. That is, a light directed in one eye results in constriction of the pupils of both eyes.

Item Description: Consensual light reflex

Question #: 13

A 25-year-old man come to the physician because of difficulty and pain when swallowing for past 12 hours. Physical examinations shows inflammation of the epiglottis. Which of the following symptoms is most likely associated with his present condition?

A. Pain that is referred to the temporomandibular joint (TMJ) joint
✓B. Otalgia (ear pain)
C. Loss of taste
D. Widened valleculae

E. Bisected the cricoid cartilage

Rationale: Otalgia because CN IX and CN X will be involved. CN V3 refers pain to TMJ joint. Inflammation of the epiglottis will lead to narrowing of valleculae.   Taste is by CN VII

Item Description: Otalgia (ear pain)

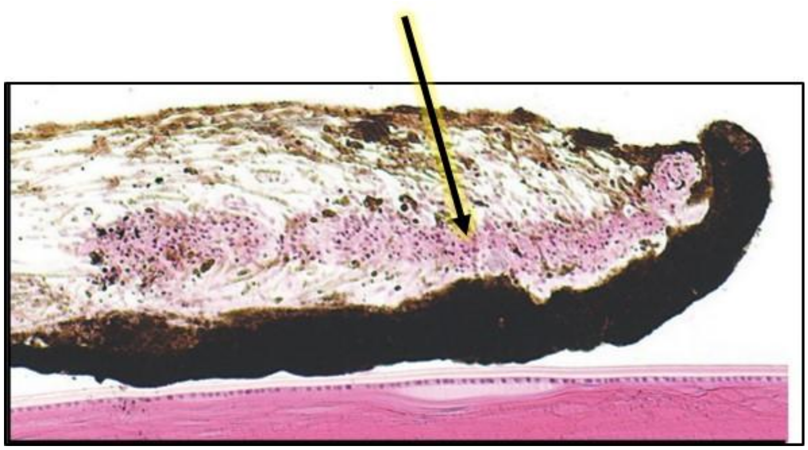
Question #: 14

A pharmaceutical company is developing novel compounds for use in eye examinations. One compound was discovered to block muscarinic acetylcholine receptors in the tissue shown in the attached photomicrograph. Which of the following would be the most likely effect of this compound?

A. Stimulates production of aqueous humor
B. Paralysis of extraocular muscles
C. Drooping of eyelids
✓D. Increases pupil size
E. Decreases pupil size

Rationale: The arrow is pointing to the constrictor pupillae muscles.   These muscles are normally responsible for constricting the pupils.   If they are inhibited by this novel compound, then their function will be affected, and the pupils will be dilated.

Attachment:



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Item Description: HCB.1340 - Iris: constrictor pupillae muscle

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Question #: 15

A 35-year-old woman comes to physician because of left dry eye and sensation of something in eye, for past six weeks. She had temporal bone fracture six weeks back and is fully recovered. Physical examination shows left eye is dry. Which of the following nerve damage is most likely leading to this condition?

A. Lesser petrosal nerve
✓B. Greater petrosal nerve
C. Tympanic nerve
D. Chorda tympani
E. Facial canal

Rationale: The greater (superficial) petrosal nerve originates at the geniculate ganglion, where the nervus intermedius and facial nerve join. It contains mainly preganglionic parasympathetic fibers and some sensory taste afferent fibers from the soft palate. After leaving the geniculate ganglion it passes through a small hiatus in the petrous bone to reach middle cranial fossa. It passes towards foramen lacerum where it is joined by the deep petrosal nerve (sympathetic fibers from the internal carotid artery) to form the nerve of the pterygoid canal (Vidian nerve). Hence supplies lacrimal gland for tearing.

Item Description: Greater petrosal nerve

Question #: 16

A 71-year-old man come to the physician because is having difficulty swallowing and often chokes on his food for the past 2-weeks. Physical examination shows absent gag reflex. Which of the following structures is most likely responsible to carry the afferent fibers in this reflex?

✓A. Cranial nerve IX
B. Cranial Nerve X
C. Cranial Nerve V
D. Cranial Nerve VII
E. Cranial Nerve XII

Rationale: The cranial nerve that provides the afferent limb of the gag reflex is the glossopharyngeal. Cranial nerve IX provides general sensation from the posterior one-third of the tongue and palatine tonsil area. When this area is touched, the soft palate and pharynx contract in a protective manner, resulting in a gag reflex. The vagus nerve is primarily responsible for the reflex's motor aspects.

Item Description: Cranial nerve IX

Question #: 17

A 64-year-old man come to physician becuase of loss of sensation on his lower lip and jaw for the past 2-months. He underwent a jaw reconstruction surgery 2 months back. Physical examination shows injury to mandibular branch of the trigeminal nerve. Which of the following additional signs and symptoms most likely be seen in this patient?

✓A. Weakness in clenching the jaw
B. Weakness in closing his upper eyelid
C. Weakness in closing his lower eyelid
D. Weakness in closing his mouth

Rationale: The trigeminal nerve's mandibular branch (CN V-3) provides sensory innervation to the skin that covers the entire length of the mandible, the mandibular teeth and gums, the lower lip and mucosa in the mouth floor, and the mucosa of the anterior 2/3 of the tongue. It also innervates the temporalis and masseter muscles, which are part of the mastication muscles that helps in clenching the lower jaw.

Item Description: trigeminal nerve's mandibular branch (CN V-3)

Question #: 18

A 28-year-old woman comes to the physician because of a 3-month history of numbness of her fingers and toes, tiredness and easy fatigability. Laboratory studies show a hemoglobin concentration of 9.2 g/dL (reference range 11.4-16.0 g/dL) and macrocytic, megaloblastic anemia. Further studies show normal serum folate levels and most of the tetrahydrofolate is in the methyltetrahydrofolate form. Which of the following laboratory findings is most likely in this patient?

A. Increased serum protoporphyrin levels
✓B. Increased serum methylmalonate levels
C. Increased urinary copper excretion
D. Low serum ferritin levels
E. Decreased red cell transketolase activity

Rationale: The vignette describes a patient with macrocytic anemia due to vitamin B12 deficiency. Note the trapping of methyltetrahydrofolate. Increased serum methylmalonate levels are characteristic of vitamin B12 deficiency. Accumulation of methylmalonate can result in demyelination which is responsible for the neurological features.

Item Description: Vitamin B12 and folic acid-quiz

Question #: 19

A 42-year-old woman comes to the physician because of a 6-month history of swelling and pain below the jaw on the right side. Physical examination shows a soft, painless mass within the submandibular gland. Excision biopsy of the mass shows normal secretory units, and multiple striated ducts. Following the removal of these ducts, which of the following functions will most likely be affected?

A. Secretion of sodium into primary saliva
B. Reabsorption of potassium from primary saliva
C. Reabsorption of bicarbonate from primary saliva
✓D. Secretion of potassium and bicarbonate into primary saliva
E. Reabsorption of potassium and bicarbonate from primary saliva

Rationale: The striated ducts are responsible for the secretion of potassium and bicarbonate into the saliva.

Item Description: HCB.1191 - Salivary glands: Submandibular

Question #: 20

A 22-year-old woman volunteers to donate oocytes for in vitro fertilization. She is started on a hormone plan to stimulate follicular growth. Subsequently, oocytes are harvested from pre-ovulatory follicles just before ovulation. These oocytes are most likely in which of the following stages of meiosis?

A. Anaphase of 2nd meiotic division
B. Prophase of 1st meiotic division
C. Prophase of 2nd meiotic division
D. Anaphase of 1st meiotic division
✓E. Metaphase of the 2nd meiotic division

Rationale: Usually the oocytes at birth are stuck in prophase of meiosis I. Just before ovulation (pre-ovulation), they complete the first meiotic division and enter the 2nd division of meiosis where they are stuck in metaphase of meiosis II until fertilization.

Item Description: HCB.1127 - Oogenesis

Question #: 21

A 6-year-old boy is brought to the physician because of a 1-week history of fever, cough and discharge from the left ear. An otoscopic examination of the left ear shows a perforation of the tympanic membrane with a purulent discharge. A diagnosis of otitis media is made. Which of the following regions of the ear is most likely affected?

A. External ear
B. Cochlear canal
C. Vestibule
✓D. Middle ear
E. Semicircular canals

Rationale: The tympanic cavity/Middle ear is an air-filled space of the middle ear connected to the nasopharynx by the auditory tube. Inflammation of the middle ear is common in children with upper respiratory infections.

Item Description: HCB.1360 - Middle ear

Question #: 22

A 38-year-old man comes to the physician because of increased thirst and frequent urination of 2 weeks duration. He has a family history of diabetes mellitus. His vital signs are all within normal limits. Physical examination shows no abnormalities. Laboratory studies show a dilute urine and decreased antidiuretic hormone levels. Following a water deprivation test, antidiuretic hormone is administered resulting in a markedly concentrated urine. Which of the following conditions does this patient most likely have?

✓A. Central diabetes insipidus
B. Nephrogenic diabetes isipidus
C. Type 1 diabetes mellitus
D. Syndrome of inappropriate diuretic hormone (SIADH)
E. Primary polydipsia

Rationale: Decreased ADH points toward central DI. The ADH challenge test confirms this suspicion because urine osmolarity increases to more normal levels after ADH administration. Accordingly, central DI is most likely.
Nephrogenic DI does not have decreased ADH and with that condition urine osmolarity should not change after the ADH challenge, hence nephrogenic DI is unlikely. Type I DM could also result in increased thirst and micturition due to glucose secretion, but that's the only symptom consistent with that diagnosis. Decreased ADH and more dilute urine are not consistent with type I DM. SIADH should come with increased ADH levels, not decreased as described in the vignette. Polydipsia also should not affect ADH levels.

Item Description: Copy of PHYS.0328 Diabetes insipidus

Question #: 23

A 25-year-old man is brought to the emergency department because of a 30-minute history of unconsciousness. Physical examination shows a depressed skull fracture. CT scan of the head shows a rapid-onset biconvex hematoma that does not cross suture lines. Which of the following diagnoses is most likely in this patient?

A. Subdural hemorrhage
B. Subarachnoid hemorrhage
C. Intracerebral hemorrhage
✓D. Epidural hemorrhage
E. Intraventricular hemorrhage

Rationale: The combination of a rapid onset, arterial origin, not crossing suture lines (due to the dura's tight adherence to the skull at sutures), and a biconvex/lenticular shape on CT are classic characteristics of an epidural hematoma. These are typically caused by trauma (like a skull fracture) rupturing meningeal arteries.

Item Description: SOM.MK.I.BPM2.2.NB.1.NSCI.0071 Intracranial hemorrhages

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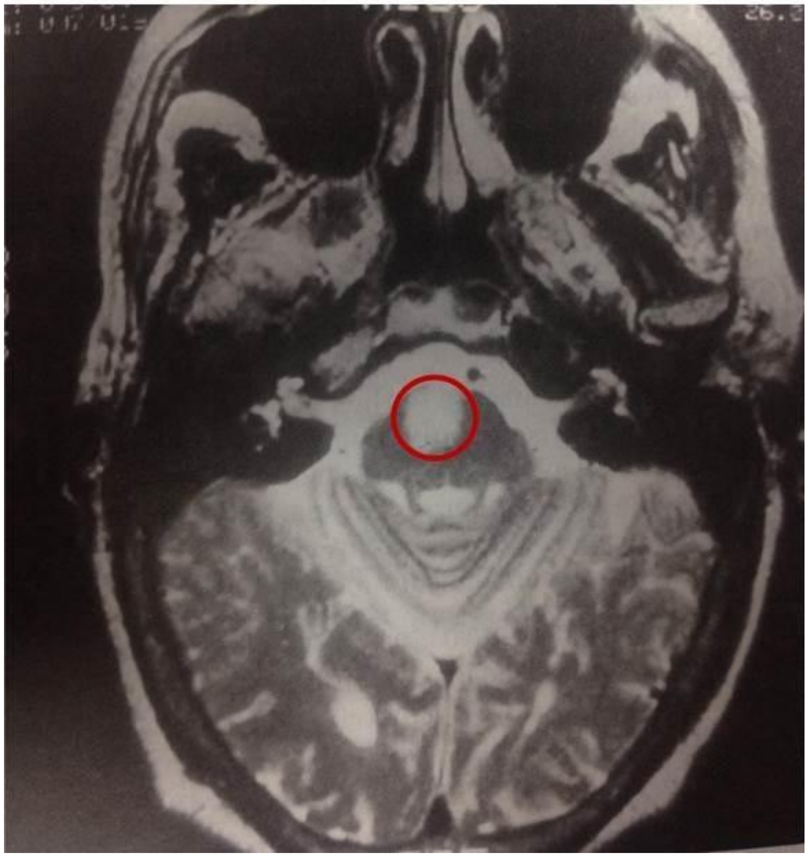
Question #: 24

A 73-year-old man is brought to the emergency department because of a 1-hour history of weakness of the limbs and trouble speaking. Imaging studies show a vascular lesion, as indicated by the encircled region in the attached image. Branches of which of the following arteries are most likely ruptured in this patient?

A. Vertebral
✓B. Basilar
C. Posterior cerebral
D. Middle cereberal
E. Anterior cerebral

Rationale: The basilar artery follows the midline of the pons in the basilar groove.
Attachment:



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Item Description: SOM.MK.I.BPM2.1.NSCI.0070 Subarachnoid vessels

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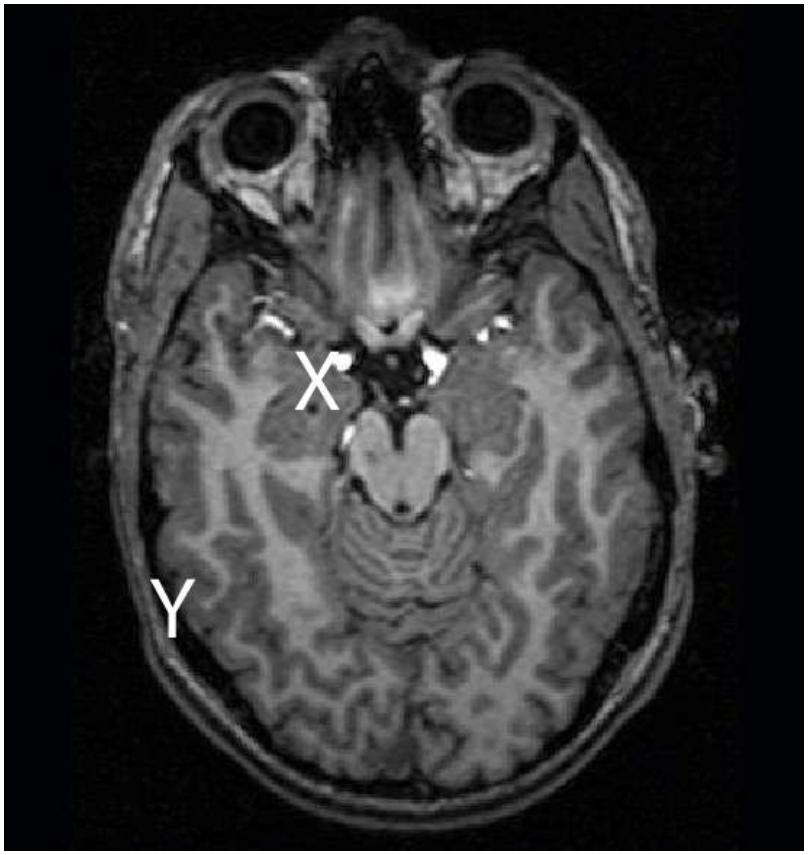
Question #: 25

A 52-year-old man is brought to the emergency department because of a 2-hour history of a traumatic head wound. Imaging studies show a fast-moving fragment of metal entered the skull at Site X and lodged at Site Y. Which of the following trajectories is most likely followed by the fragment as it passes through the brain?

A. Anteromedial
B. Posteromedial
C. Anterolateral
✓D. Posterolateral
E. Dorsomedial
F. Ventromedial
G. Dorsolateral
H. Ventrolateral

Rationale: Both Y and X correspond to the telencephalon. Y lies posterior to X (X is closer to the face). Y also lies farther from the midline than does Y. Hence, the fragment has traveled posterolaterally.

Attachment:



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Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0009 Neuroimaging anatomical vectors

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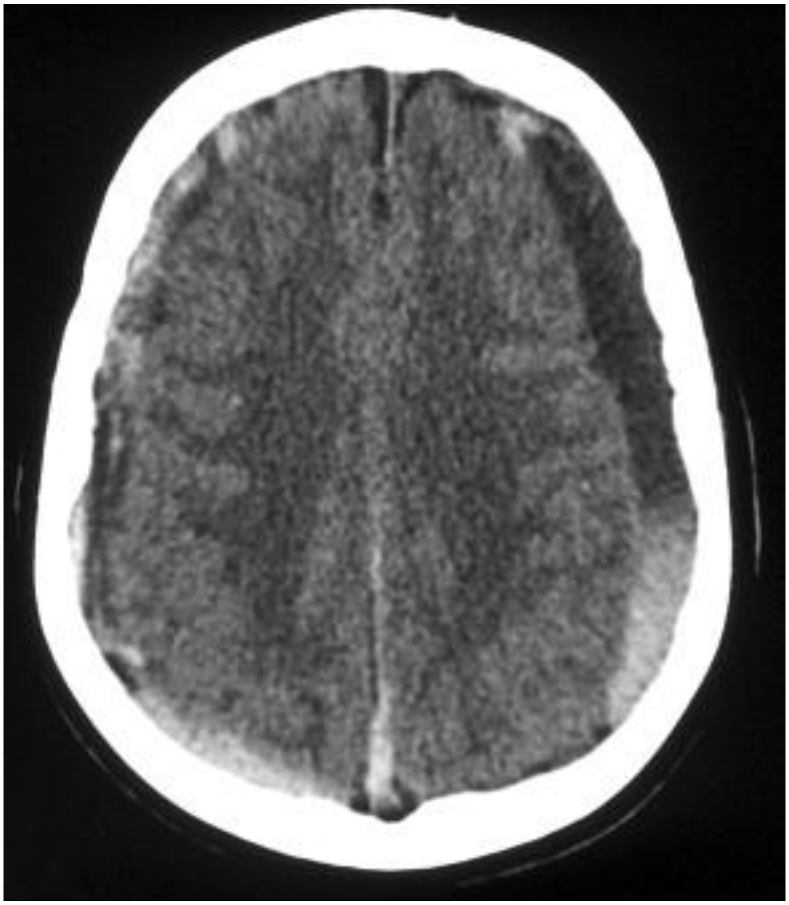
Question #: 26

A 14-year-old boy is brought to the emergency department by his parents because of a 5-hour history of irritability, intense headache, nausea and vomiting that occurred following a blow to the head. Imaging studies show hematoma (see attached). Which of the following hemorrhagic diagnoses is most likely?

A. Left epidural
✓B. Left subdural
C. Left subarachnoid
D. Left periventricular
E. Right epidural
F. Right subdural
G. Right subarachnoid
H. Right periventricular

Rationale: A subdural hemorrhage is indicated by the crescent-shaped accumulation of blood that crosses suture lines. The clinical presentation is also consistent with, but not limited to, a subdural hemorrhage.

Attachment:



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Item Description: SOM.MK.I.BPM2.2.NB.1.NSCI.0071 Distinguish hemorrhages

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Question #: 27

A 48-year-old man comes to the physician because of a 4-hour history of nausea, diarrhea and stomach cramps. He says he had accidentally ingested some household insecticide shortly before experiencing gastric discomfort. Which of the following substances is most likely affected by the toxin?

✓A. Acetylcholine esterase
B. Dopa decarboxylase
C. 5-hydroxyindolacetic acid
D. Tyrosine hydroxylase
E. Glutamic acid decarboxylase

Rationale: The gastic distress likely affects overactivation of the cholinergic part of the parasympathetic nervous system. Insecticides commonly suppress acetylcholine esterase, thereby suppressing metabolism of ACh. Other peripheral manifestations of cholinergic overactivation are likely (e.g., sweating).

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0105 Acetylcholine synthesis-removal

Question #: 28

A 24-year-old woman comes to the physician because of a 2-hour history of penetrating injuries affecting her left leg. A local anesthetic is applied via the subarachnoid space for pain relief and suppression of movement of the lower limbs. Which neuronal type and ion channel are most likely linked to the paralytic effects of the drug?

A. Inhibitory interneuron; calcium
B. Inhibitory interneuron; sodium
C. Inhibitory interneuron; chloride
D. Inhibitory interneuron; magnesium
E. Inhibitory interneuron; potassium
F. Motor neuron; calcium
✓G. Motor neuron; sodium
H. Motor neuron; chloride
I. Motor neuron; magnesium
J. Motor neuron; potassium

Rationale: Local anesthetics like lidocaine block voltage-gated Na channels (NaV), inhibiting electrical excitability. Such an effect on inhibitory interneurons in the spinal cord would be enhancing muscle excitability, hence not leading to paralysis. Paralysis may result when NaV channels on motor neurons or myocytes are blocked, making (g) the correct answer.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0035 Lidocaine action potential

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Question #: 29

A 47-year old man is brought to the emergency department because of a 3-hour history of nausea and a fainting spell. Imaging studies show hypoperfusion in a small localized area of the brain. An infarction is suspected. Which of the following cells will most likely proliferate to form a scar in the injured area?

A. Schwann cells
B. Microglia
C. Oligodendrocytes
✓D. Astrocytes
E. Ependymocytes

Rationale: Astrocytes are the glial cells that form scars in the central nervous system.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0017 Glial cells

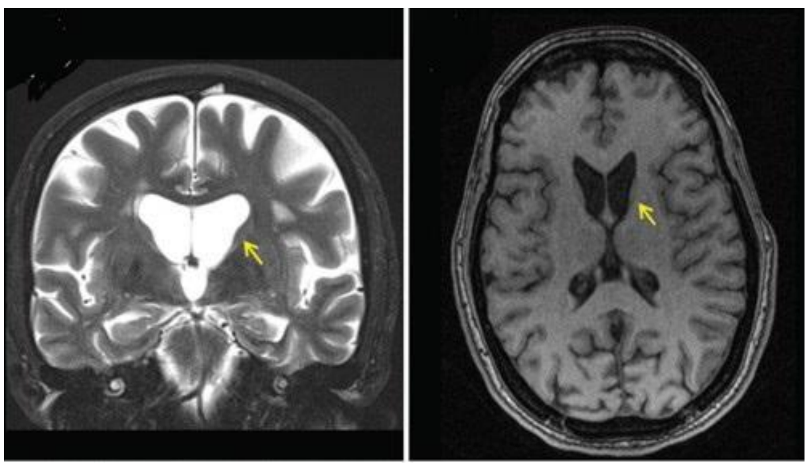
Question #: 30

A 38-year-old man comes to the physician because of a 6-week history of involuntary jerking movements of the hands and face. He says that he noticed twitching in his fingers 2-months ago. Family history shows that his father had died 6 years after developing similar signs. Imaging studies show enlargement of the lateral ventricles secondary to bilateral atrophy of the caudate nuclei (yellow arrow in attachment). Which of the following forms of hydrocephalus is most likely affecting the patient?

A. Obstructive
B. Communicating
C. Normal pressure
✓D. Ex vacuo
E. Non-communicating

Rationale: Hydrocephalus arising secondary to degeneration of brain tissue is ex vacuo. The patient may be expressing hydrocephalus ex vacuo secondary to neurodegeneration associated with Huntington disease.

Attachment:



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Item Description: SOM.MK.II.BPM2.1.NB.2.NSCI.0067 Hydrocephalus causes